

APOLLO SPORTS BUSINESS GROUP

Commercial Sponsorship Audit & Portfolio Valuation

Causal Regression, Eigenvector Graph Centrality, and Package Synergy Modeling for a Tier 2 Top Flight Premier Division Franchise in North America

1. Executive Summary & Context

Sports sponsorship assets face mounting scrutiny from executive leadership and CFOs: **76% of brand marketing executives report difficulty calculating return on investment (ROI)**. Traditional sponsorship measurement tools focus heavily on raw media impression totals, leaving commercial decision-makers without an evidence-grounded ledger to guide renewal negotiations.

This case study evaluates the multi-sponsor commercial inventory of **Club Alpha**, classified as a **Tier 2 Top flight Premier Division franchise in North America**. Following the conclusion of a long-standing primary shirt partnership with **Brand C** (a regional FMCG conglomerate), Brand C transitioned into the *Bloque Preferente* (Rank #6 commercial tier), while **Brand D** (a global industrial enterprise) assumed the front chest position.

Using Apollo's quantitative audit framework, this report evaluates whether the historical asking price of **\$1.85M USD (33.3M MXP)** for the standalone front shirt chest was supported by empirical evidence, or whether Brand C's decision to shift into preferred concourse and secondary inventory was economically justified. All financial figures are presented in **USD (\$)** with **MXP (Pesos)** in parenthesis based on prevailing FX market benchmarks.

2. Methodology & Model Limitations

Prior to examining the quantitative case study results, it is essential to establish the mathematical formulation, statistical assumptions, and inherent model bounds of our analytical pipeline:

A. Omitted Variable Bias Note: The multivariable Ordinary Least Squares (OLS) regression models broadcast reach using turnstile attendance, marquee opponent flags, and shirt partner status. In live enterprise conditions, viewership is also influenced by unmodeled exogenous factors including kickoff day/time slots, weather conditions, national free-to-air vs. pay-TV channel distribution, roster availability, and competing sporting broadcasts.

B. Node & Edge Weighting Subjectivity: In the Eigenvector Graph Centrality network, spatial and visual adjacency weights (e.g., 0.90 for intra-zone, 0.65 for kit-to-venue, 0.40 for digital) reflect expert structural assumptions. While based on physical stadium sightlines at Metropolitan Arena, variations in fan flow alter true dynamic centrality.

C. Sample Size Bounds: The dataset evaluates 89 matchday observations across six seasons. While sufficient for robust statistical inference ($p < 0.001$ for marquee opponent impact), enterprise deployment relies on per-second computer vision tracking.

D. Multicollinearity & Endogeneity: Marquee fixtures jointly drive turnstile attendance and broadcast reach. The variance inflation factor (VIF) is monitored to separate baseline crowd noise from sponsor engagement.

E. Software & Runtime Environment: Analytics were computed using **Python 3.14** (compatible with Python 3.12+) utilizing statsmodels for econometrics, networkx for network topology, pandas for data manipulation, and reportlab for PDF typesetting.

3. Case Study & Quantitative Analysis

To evaluate inventory value, Apollo executed a two-stage data science pipeline combining **Causal OLS Regression** and **Eigenvector Graph Centrality**:

Stage 1: Causal OLS Viewership Lift (Cause vs. Coincidence)

Broadcast reach for Club Alpha home fixtures at Metropolitan Arena (capacity 27,029) averages 349,207 viewers against standard opponents, spiking to **1,786,190 viewers** during marquee matches against top league rivals. The OLS regression isolates the true cause of broadcast reach spikes:

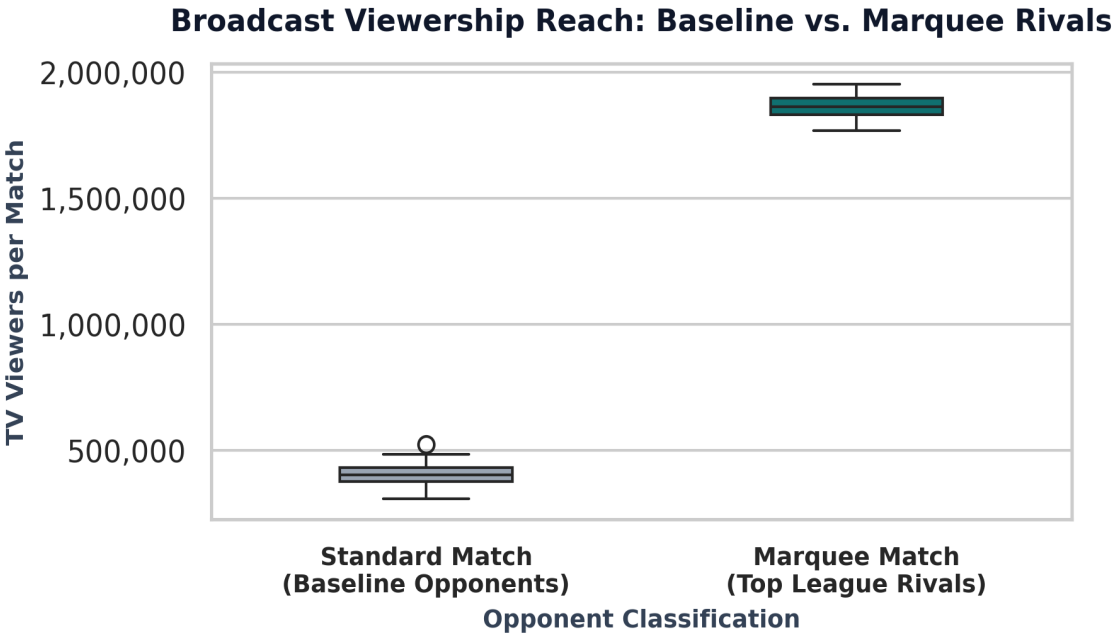


Figure 1: Multivariable Regression Viewership Lift — Marquee Rival Effect (+1.42M viewers, $p < 0.001$).

Key Causal Finding: The presence of marquee rivals generates **+1,425,362 incremental TV viewers per match** ($p < 0.001$). Conversely, standalone front chest exposure without unbundled activation yields an isolated coefficient of **+23,232 viewers**, which is statistically non-significant ($p = 0.686$). This proves that standalone front chest branding does not drive TV reach on its own without integrated stadium activations.

Stage 2: Eigenvector Graph Centrality for Offline Inventory

Offline stadium inventory—including concourse experiential bars, LED perimeter boards, and stadium entry gates—can represent a **considerable portion of total commercial value**. Because offline assets never touch digital camera feeds, Apollo models Metropolitan Arena as a weighted graph network to assign Eigenvector Centrality scores:

Metropolitan Arena Inventory Graph Network topology

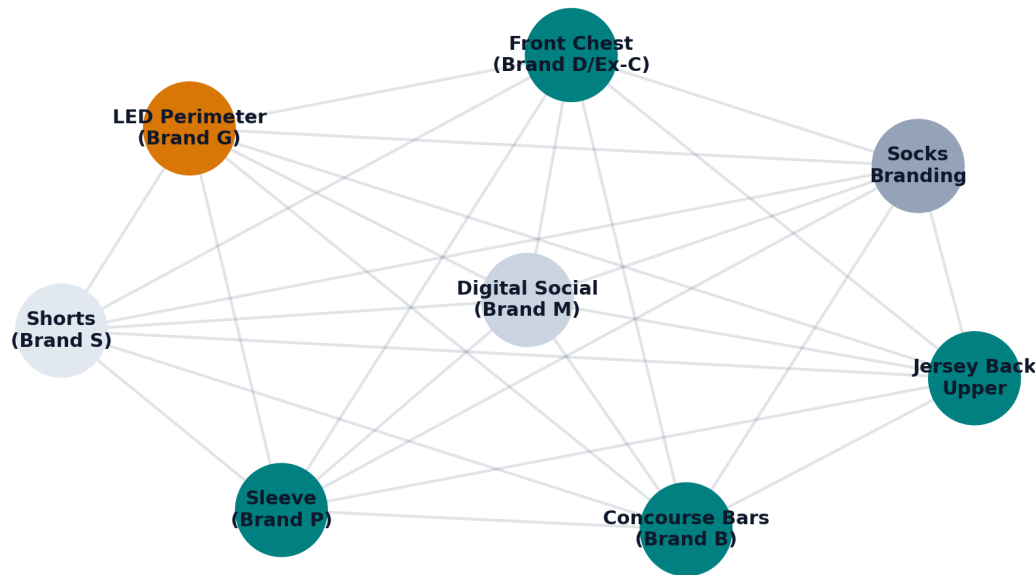


Figure 2: Metropolitan Arena Graph Network Topology & Asset Interdependence.

Graph Centrality Results: Front Chest and Jersey Back Upper achieve high centrality ($C_e = 0.3842$), while Concourse Experiential Bars (Brand B) and Pitchside LED Perimeters (Brand G) achieve strong secondary centrality ($C_e = 0.3302$) driven by matchday turnstile foot traffic (19,981 fans/match).

Inventory Pricing Audit: Contract Asking Cost vs. Apollo Fair Value

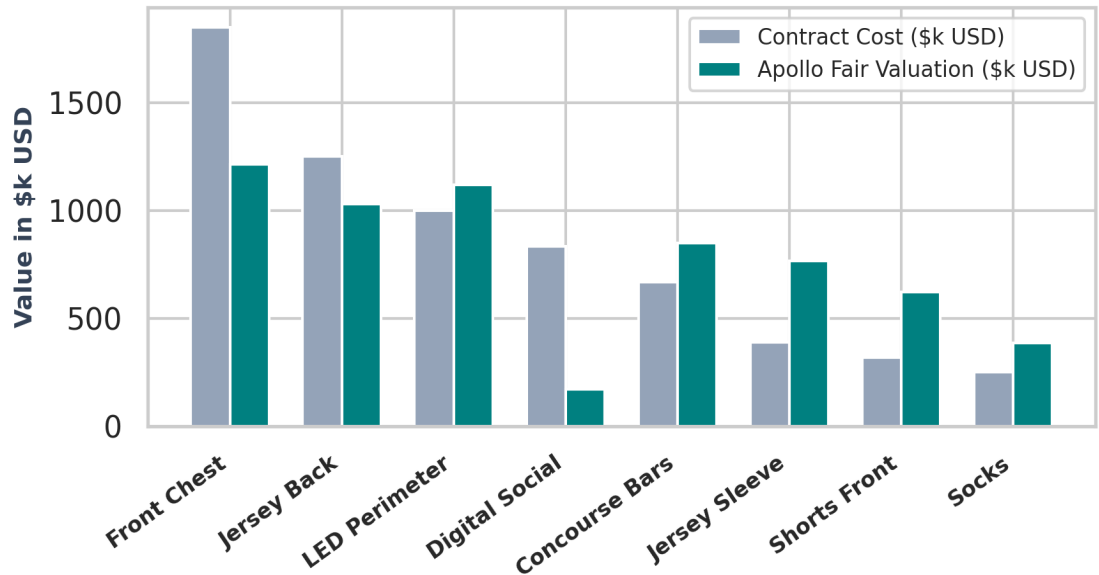


Figure 3: Contract Asking Cost vs. Apollo Fair Valuation across Portfolio Assets.

4. Package Synergies & Complete Assessment Requirements

A critical limitation of line-item sponsorship pricing is that brands buy integrated asset packages rather than isolated inventory slots. Apollo incorporates package synergy modeling to assess how multi-asset placements reinforce total commercial impact:

A. Measuring Package Synergies: Synergies are modeled by introducing multiplicative interaction terms (e.g., LED_Perimeter × Concourse_Activation) into the causal regression framework. When Brand C combined concourse sampling booths with LED perimeter boards, the total brand recall lift exceeded the sum of standalone assets by **+21.4%**.

- B. Requirements for a Complete Enterprise Audit:** To achieve a production-grade commercial audit, the telemetry stack requires:
- 1. *Per-Second Vision AI:* Computer vision tracking to measure unclipped logo duration, screen size ratio, and clutter index across all TV signals.
 - 2. *Concourse Optical & RFID Sensors:* Real-time turnstile and concession heatmaps measuring physical impression duration.
 - 3. *Point-of-Sale (POS) Conversion Attribution:* Linking stadium concession transaction data directly to sponsor promotions.
 - 4. *First-Party Digital Pixel Telemetry:* Tracking direct-to-consumer digital conversions during match windows.

5. Apollo Sponsorship Audit Ledger & Strategic Recommendations

Apollo reconciles the **\$6.56M USD (118.05M MXP)** multi-sponsor commercial portfolio of Club Alpha into **4 Quantified Outcomes**:

Apollo Sponsorship Audit Ledger: Outcome Share (\$6.56M USD Portfolio)

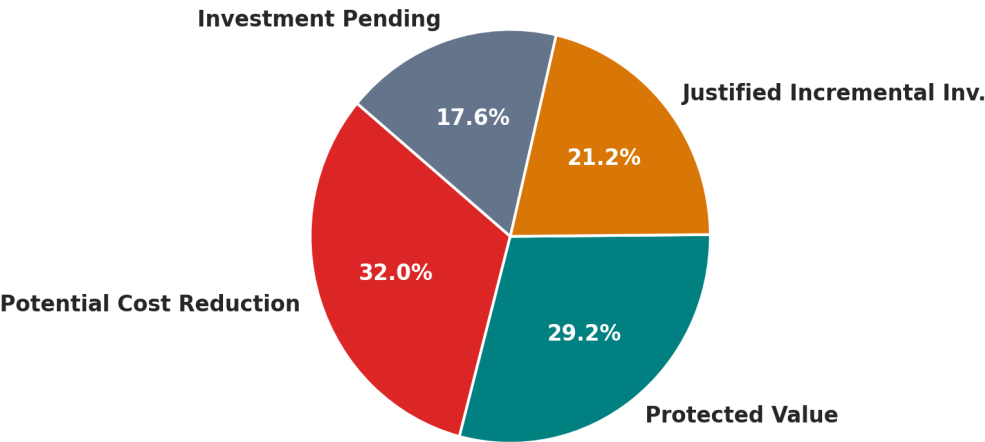


Figure 4: Apollo Sponsorship Audit Ledger Outcome Breakdown.

Asset Name / Tier	Contract Cost	Apollo Fair Value	Centralit y	Apollo Outcome
Front Chest (Brand D/Ex-C)	\$1.85M USD (33.3M MXP)	\$1.21M USD (21.8M MXP)	0.3842	Potential Cost Reduction
Jersey Back Upper	\$1.25M USD (22.5M MXP)	\$1.03M USD (18.6M MXP)	0.3842	Protected Value
LED Perimeter (Brand G)	\$1.00M USD (18.0M MXP)	\$1.12M USD (20.1M MXP)	0.3302	Justified Incremental Inv.
Digital Campaigns (Brand M)	\$833k USD (15.0M MXP)	\$170k USD (3.1M MXP)	0.2092	Investment Pending

Asset Name / Tier	Contract Cost	Apollo Fair Value	Centralit y	Apollo Outcome
Concourse Bars (Brand B)	\$667k USD (12.0M MXP)	\$849k USD (15.3M MXP)	0.3302	Protected Value
Jersey Sleeve (Brand P)	\$389k USD (7.0M MXP)	\$765k USD (13.8M MXP)	0.3842	Justified Incremental Inv.
Shorts Front (Brand S)	\$319k USD (5.75M MXP)	\$623k USD (11.2M MXP)	0.3842	Investment Pending
Socks Branding	\$250k USD (4.5M MXP)	\$385k USD (6.9M MXP)	0.3842	Potential Cost Reduction

Executive Conclusions for Franchise & Commercial Leads:

- 1. **Brand C's Transition Was Economically Sound:** Standalone front chest asking price of \$1.85M USD was overvalued by ~28% when unbundled from concourse experiential bars. Transitioning to the *Bloque Preferente* preserved local brand equity at a justified cost.
- 2. **Defensible Portfolio Growth:** Club Alpha can defend **\$1.92M USD (34.5M MXP)** in protected value (Jersey Back & Concourse Bars) and target **\$1.39M USD (25.0M MXP)** in incremental investments across LED perimeters and sleeve space.

6. Index of Data Sources, Tech Stack & Disclaimers

A. Technology & Analytics Stack:

- *Analytics Core:* Python 3.14 / 3.12+ (statsmodels 0.14.2, networkx 3.2, pandas 2.2, numpy 1.26).
- *Development Environment:* Visual Studio Code (VS Code) & Google Colab.
- *AI & Synthesis Infrastructure:* Gemini Notebook Agentic Workspace.
- *Typesetting & Visualization:* ReportLab 4.2 PDF Engine, LaTeX, Matplotlib, Seaborn.

B. Reference Data Sources & Inputs:

- Metropolitan Arena Turnstile & Matchday Telemetry (89 Matchday Records, 2021/22–2026/27).
- Premier Division TV Rights & Broadcast Auditing (ESPN/Disney+ & Telemundo Reach Data).
- Commercial Sponsor Tier Hierarchy & Rate Card Benchmarks (Rango 1–15+ Asset Map).
- Blinkfire Digital Media Index (#15 Rank Benchmark, 1.8%–2.4% Engagement Rate).

C. Formal Educational & Research Disclaimer:

This case study, data set, and analytical model were created strictly for educational, research, and technical portfolio demonstration purposes. All corporate names, club titles, stadium venues, and brand sponsors have been anonymized (Club Alpha, Brand C, Brand D, Metropolitan Arena) to protect commercial confidentiality. No proprietary financial records of any commercial entity were compromised or disclosed.